

12.224

EE25BTECH11002 - Achat Parth Kalpesh

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Question

The lower triangular matrix $\mathbf{L}$ in the LU factorization of the matrix $\begin{pmatrix} 25 & 5 & 4 \\ 10 & 8 & 16 \\ 8 & 10 & 22 \end{pmatrix}$ is written as $\begin{pmatrix} 1 & 0 & 0 \\ L_{21} & 1 & 0 \\ L_{31} & L_{32} & 1 \end{pmatrix}$. The element L_{32} is

1 1.0

2 1.4

3 0.4

4 0.32

Solution

Let

$$\mathbf{A} = \begin{pmatrix} 25 & 5 & 4 \\ 10 & 8 & 16 \\ 8 & 10 & 22 \end{pmatrix} \quad (1)$$

The lower triangular and upper triangular matrices are written as

$$\mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ L_{21} & 1 & 0 \\ L_{31} & L_{32} & 1 \end{pmatrix}, \quad \mathbf{U} = \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{pmatrix} \quad (2)$$

and

$$\mathbf{A} = \mathbf{LU} \quad (3)$$

$$a_{ij} = \sum_{k=1}^{\min(i,j)} L_{ik} U_{kj} \quad (4)$$

Using Doolittle-style relations (and the elementwise formula):

$$u_{11} = a_{11} = 25 \quad (5)$$

$$u_{12} = a_{12} = 5 \quad u_{13} = a_{13} = 4 \quad (6)$$

$$L_{21} = \frac{a_{21}}{u_{11}} = \frac{10}{25} = 0.4 \quad (7)$$

$$L_{31} = \frac{a_{31}}{u_{11}} = \frac{8}{25} = 0.32 \quad (8)$$

$$u_{22} = a_{22} - L_{21}u_{12} = 8 - (0.4)(5) = 8 - 2 = 6 \quad (9)$$

$$u_{23} = a_{23} - L_{21}u_{13} = 16 - (0.4)(4) = 16 - 1.6 = 14.4 \quad (10)$$

Solution

Now use the formula for a_{32} (with $\min(3, 2) = 2$)

$$a_{32} = L_{31}u_{12} + L_{32}u_{22} \quad (11)$$

Solve for L_{32} :

$$L_{32} = \frac{a_{32} - L_{31}u_{12}}{u_{22}} = \frac{10 - (0.32)(5)}{6} = \frac{10 - 1.6}{6} = \frac{8.4}{6} = 1.4 \quad (12)$$

$$L_{32} = 1.4 \quad (13)$$